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Lab 3: Architectural Pattern Justification

System: Database Query Performance Profiler

Architecture Selection: *We chose Layered Architecture for the Database Query Performance Profiler System.*

Architectural Choice

The system is organized into three horizontal layers: a Presentation Layer (Query Interface Component), a Business/Application Layer (Query Execution Monitor, Performance Analyzer, and Report Generator Components), and a Data Layer (Database Connector and Metrics Storage Components). Each layer only communicates with the layer directly adjacent to it through well-defined provided/required interfaces.

Two Reasons for This Choice

- **Sequential, well-defined workflow:** Profiling a query naturally follows a fixed pipeline — submit query, capture execution statistics, analyse performance metrics, then generate a report. This linear flow maps directly onto layers, so a layered structure keeps the design simple and easy for a small team to build, test, and maintain, unlike a distributed style that would add unnecessary operational overhead for a single analytical tool.
- **Isolation of volatile components:** The scoring logic used by the Performance Analyzer (thresholds, weighting of slow-query indicators) is likely to change often as the tool matures, while the presentation and data-access concerns change rarely. Layering isolates that volatility inside the Business Layer so the UI and the database/storage access code do not need to be touched when the analysis rules evolve.

Security Advantage

By confining all database connection strings and credentials to the Database Connector Component in the Data Layer, the Presentation Layer and end users never have direct access to raw database access details. All query execution requests must pass through the Query Execution Monitor's required interface, giving the system a single, centralised point at which to enforce authentication, input validation, and audit logging before any SQL reaches the database..

Performance Benefit

Because the Metrics Storage Component sits behind a dedicated interface in the Data Layer, the Performance Analyzer can cache recent execution metrics and historical trend data in the Business Layer rather than re-querying the database for every report. This reduces redundant round-trips to the profiled database, which matters because the profiler must add minimal overhead to the very queries it is measuring.

